

OVERVIEW OF THE HINTS 5 CYCLE 4 (2020) SURVEY AND DATA ANALYSIS RECOMMENDATIONS

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CONTENTS

Overview of HINTS 3
HINTS 5 3
Methodology 3
Sample Size and Response Rates 3
Analyzing HINTS Data 4
Important Analytic Variables in the Database 5
Variance Estimation Methods: Replicate vs. Taylor Linearization 6
Denominator Degrees of Freedom (DDF) 7
Statistical Software Example Code 8
Analyzing Data Using SAS 8
Analyzing Data Using SPSS—Taylor Series 18
Analyzing Data Using Stata 27
Merging HINTS Survey Iterations 46
Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using SAS 46
Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using SPSS 49
Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using Stata 51
References 54

Overview of HINTS

The Health Information National Trends Survey (HINTS) is a nationally representative survey that has been administered every few years by the National Cancer Institute since 2003. The HINTS target population is all adults aged 18 or older in the civilian non-institutionalized population of the United States. The HINTS program collects data on the American public's need for, access to, and use of health-related information and health-related behaviors, perceptions, and knowledge. (Hesse, et al., 2006; Nelson, et al., 2004). Previous iterations include HINTS 1 (2003), HINTS 2 (2005), HINTS 3 (2007/2008), HINTS 4, Cycle 1 (2011); HINTS 4, Cycle 2 (2012); HINTS 4, Cycle 3 (2013); HINTS 4, Cycle 4 (2014); HINTS-FDA, Cycle 1 (2015); HINTS-FDA, Cycle 2 (2017); HINTS 5, Cycle 1 (2017); HINTS 5, Cycle 2 (2018), and HINTS 5, Cycle 3 (2019).

HINTS 5

The HINTS 5 administration includes four data collection cycles over four years, starting in 2017. The first of these cycles (HINTS 5, Cycle 1) was conducted from January through May 2017. The focus of this report is HINTS 5, Cycle 4, collected February through June 2020. HINTS 5 Cycle 4 draws upon the lessons learned from prior iterations of HINTS but did not include a web option like HINTS 5 Cycle 3 (2019). A single-mode mail survey was implemented for HINTS 5, Cycle 4 (2020). For more extensive background about the HINTS program and previous data collection efforts, see Finney Rutten, et al. (2012).

Methodology

Data collection for Cycle 4 of HINTS 5 began in February 2020 and concluded in June 2020. HINTS 5, Cycle 4 was a self-administered mailed questionnaire, using a sampling frame provided by Marketing Systems Group (MSG) of addresses in the United States. All addresses were grouped into two strata: 1) addresses in areas with high concentrations of minority populations; 2) addresses in areas with low concentrations of minority populations. All non-vacant residential addresses in the United States present in the MSG database, including post office (P.O.) boxes, throwbacks (i.e., street addresses for which mail is redirected by the U.S. Postal Service to a specified P.O. box), and seasonal addresses were subject to sampling. However, P.O. box addresses were only used if they were classified as the only-way-to-get mail. The protocol for mailing the questionnaires involved an initial mailing of the questionnaire, followed by a reminder postcard, and up to two additional mailings of the questionnaire as needed for non-responding households. All households received one English survey per mailing unless someone from the household contacted Westat to request a Spanish survey, in which case the household received one Spanish survey per mailing for all subsequent mailings. The second stage of sampling consisted of selecting one adult within each sampled household using the next-birthday method. In this method, the adult who would have the next birthday in the sampled household was asked to complete the questionnaire. A $2 monetary incentive was included with the survey to encourage participation. Refer to the HINTS 5, Cycle 4 Methodology Report for more extensive information about the sampling procedures.

Sample Size and Response Rates

The final HINTS 5, Cycle 4 (2020) sample consists of 3,865 respondents. Note that 73 of these respondents were considered partial completers who did not answer the entire survey. To calculate the completion rate, questions required of every respondent were assessed; a questionnaire was considered to be complete if at least 80% of these required questions were answered. A questionnaire was considered to be partially complete if 50%–79% of the questions were answered in Sections A and B. There were 25 questionnaires with fewer than 50% of the required questions in A and B answered. Those questionnaires were discarded. Household response rates were calculated using the American Association for Public Opinion Research response rate 4 (RR4) formula. The overall household response rate using the next-birthday method was 36.7%. See the Methodology Report for more information.

Analyzing HINTS Data

If you are solely interested in calculating point estimates (means, proportions, etc.), either weighted or unweighted, you can use programs including SAS, SPSS, Stata, and Systat. If you plan on doing inferential statistical testing using the data (i.e., anything that involves calculating a p-value or confidence interval), it is important that you utilize a statistical program that can compute the correct variance estimates when analyzing survey data that employ a complex sampling method, such as employed for HINTS. The issue is that the standard errors in your analyses will most likely be underestimated if you do not take into account the sampling procedure; therefore, your p-values will be smaller than they "should" be, your tests will be more liberal, and you are more likely to make a type I error. HINTS data contain jackknife replicate weights to compute the correct variance estimates. Statistical programs like SUDAAN, Stata, SAS, R, and Mplus can incorporate the replicate weights found in the HINTS database.

In addition, with HINTS 5, Cycle 4 (2020) all datasets will contain variance codes that will allow for inferential statistical testing using Taylor Series Linearization. This can be used along with the Complex Samples module found in SPSS or with other statistical programs that support using Taylor Linearization. Please see the "Important Analytic Variables in the Database" section for more information about the variance codes, and the "Variance Estimation Methods: Replicate vs. Taylor Linearization" section for more information about the two variance estimation methods.

Note that analyses of HINTS variables that contain a large number of valid responses usually produce reliable estimates, but analyses of variables with a small number of valid responses may yield unreliable estimates, as indicated by their large variances. The analyst should pay attention to the standard error and coefficient of variation (relative standard error) for estimates of means, proportions, and totals, and the analyst should report these when writing up results. It is important that the analyst realizes that small sample sizes for certain analyses will tend to result in unprecise estimates. Methods for obtaining confidence intervals for small proportions or limited degrees of freedom for small populations are described in Korn and Graubard's Analysis of Health Surveys (1999; pp. 64-68). Related to this, beginning with HINTS 5 Cycle 4 (2020), the HINTS program has implemented data suppression thresholds wherein some variables with cells that have <25 responses are either collapsed, recoded to missing, or deleted/suppressed entirely. Thresholds were determined based solely on respondent disclosure risk, but small cell sizes also have implications for precision. Please see the Methodology Report for more information, including information on which variables were recoded or suppressed.

HINTS 5 Cycle 4 (2020) included a novel item. Item P15 (PoliticalViewpoint) asked "Thinking about politics these days, how would you describe your own political viewpoint?" and included a 7-point response scale representing a continuum from very liberal to very conservative. This item was included because HINTS 5 Cycle 4 also included new items to assess support for and opposition to communication-related public health policies such as product advertising restrictions and warning labels. HINTS analyses that use modeling to assess the adjusted prevalence of, or the predicted probability of, policy support or opposition should control for political viewpoint so that models are not mis-specified, as the literature demonstrates that most of the variance in policy support outcomes is explained by political ideology or party identification. Excluding the political viewpoint variable from these analyses could lead to overestimating the contribution of other variables in predictive models. Aside from using the variable with policy-related outcomes or outcomes assessing trust in information sources for health-related information, the HINTS program strongly recommends that careful consideration be given before using political viewpoint as an independent variable in non-policy-related studies. The political viewpoint variable used on HINTS 5 Cycle 4 came from the 2012 Cooperative Congressional Election Study (CCES), and a very similar question and 7-point scale is used in the General Social Survey (GSS) Ballot 1 2015, and in the American National Election Study (ANES) 2016 Time Series Study. (Political party [republican, democrat, independent, other] was asked in the ASCO National Cancer Opinion Survey 2018 but was not asked on HINTS.)

Important Analytic Variables in the Database

Refer to the HINTS 5, Cycle 4 Methodology Report for more information regarding the weighting and stratification variables listed below.

Note that estimates from the 2018 American Community Survey (ACS) of the U.S. Census Bureau were used to calibrate the HINTS 5, Cycle 4 control totals with the following variables: age, gender, education, marital status, race, ethnicity, and census region. In addition, variables from the 2017 National Health Interview Survey (NHIS) were used to calibrate HINTS 5, Cycle 4 data control totals regarding: percent with health insurance and percent ever had cancer.

Final Sample and Replicate Weights for Jackknife Replication

PERSON_FINWT0: Final sample weight used to calculate population point estimates.

PERSON_FINWT1 through PERSON_FINTW50: Fifty replicate weights that can be used to calculate accurate standard error of estimates using the jackknife replication method. More information about how these weights were created can be found in the "HINTS 5, Cycle 4 Methodology Report" included in the data download or see Korn and Graubard (1999).

Stratum/Cluster Variables and Final Sample Weights for Taylor Series Linearization Methods

VAR_STRATUM: This variable identifies the first-stage sampling stratum of a HINTS sample for a given data collection cycle. It is the variable assigned to the STRATA parameter when specifying the sample design to compute variances using the Taylor Series linearization method. It has two values: high minority (HM) and low minority (LM).

VAR_CLUSTER: This variable identifies the cluster of sampling units of a HINTS sample for a given data collection cycle used for estimating variances. It is the variable assigned to the CLUSTER parameter when specifying the sample design to compute variances using the Taylor Series linearization method. It has values ranging from 1 to 50.

Other Variables

BirthGender: New with H5C4, the question wording was changed to reflect new guidelines for assessing gender. The question wording read, "On your original birth certificate, were you listed as male or female?" The new variable name reflects this change in question wording.

STRATUM: This variable codes for whether the respondent was in the Low or High Minority Area sampling stratum.

APP_REGION: This variable codes for Appalachia subregion.

LANGUAGE_FLAG: This variable codes for the language the survey was completed in (English or Spanish).

INCOMERANGES_IMP: This is the income variable (INCOMERANGES) imputed for missing data. To impute for missing items, PROC HOTDECK from the SUDAAN statistical software was used. PROC HOTDECK uses the Cox-Iannacchione Weighted Sequential Hot Deck imputation method, as described by Cox (1980). The following variables were used as imputation classes given their strong association with the income variable: Education (O3), Race/Ethnicity (RaceEthn) (standard recode from O5 and O6), and Occupational Status (FullTimeOcc_Cat).

Variance Estimation Methods: Replicate vs. Taylor Linearization

Variance estimation procedures have been developed to account for complex sample designs. Taylor series (linear approximation) and replication (including jackknife and balanced repeated replication, BRR) are the most widely used approaches for variance estimation. Either of these techniques allow the analyst to appropriately reflect factors such as the selection of the sample, differential sampling rates to subsample a subpopulation, and nonresponse adjustments in estimating sampling error of survey statistics. Both procedures have good large sample statistical properties, and under most conditions, these procedures are statistically equivalent. Wolter (2007) is a useful reference on the theory and applications of these methods.

The HINTS 5, Cycle 4 datasets include variance codes and replicate weights so analysts can use either Taylor Series or replication methods for variance estimation. The following points may provide some guidance regarding which method will best reflect the HINTS sample design in your analysis.

TAYLOR SERIES	REPLICATION METHODS
• Most appropriate for simple statistics, such as means and proportions, since the approach linearizes the estimator of a statistic and then uses standard variance estimation methods.	• Useful for simple statistics such as means and proportions, as well as nonlinear functions. • Easy to use with a large number of variables. • Better accounts for variance reduction procedures such as raking and post-stratification. However, the variance reduction obtained with these procedures depends on the type of statistic and the correlation between the item of interest and the dimensions used in raking and post-stratification. Depending on your analysis, this may or may not be an advantage.

The Taylor Series variance estimation procedure is based on a mathematical approach that linearizes the estimator of a statistic using a Taylor Series expansion and then uses standard variance methods to estimate the variance of the linearized statistic.

The replication procedure, on the other hand, is based on a repeated sampling approach. The procedure uses estimators computed on subsets of the sample, where subsets are selected in a way that reflect the sample design. By providing weights for each subset of the sample, called replicate weights, end users can estimate the variance of a variety of estimators using standard weighted sums. The variability among the replicates is used to estimate the sampling variance of the point estimator.

An important advantage of replication is that it provides a simple way to account for adjustments made in weighting, particularly those with variance-reducing properties, such as weight calibration procedures. (See Kott, 2009, for a discussion of calibration methods, including raking, and their effects on variance estimation). The survey weights for HINTS were raked to control totals in the final step of the weighting process. However, the magnitude of the reduction generally depends on the type of estimate (i.e., total, proportion) and the correlation between the variable being analyzed and the dimensions used in raking.

Although SPSS's estimates of variance based on linearization take into account the sample design of the survey, they do not properly reflect the variance reduction due to raking. Thus, when comparing across Taylor series and replicate methods, analyses with Taylor series tend to have larger standard errors and generally provide more conservative tests of significance. The difference in the magnitude of standard errors between the two methods, however, will be smaller when using analysis variables that have little to no relationship with the raking variables.

Denominator Degrees of Freedom (DDF)

Replicate Weights: The HINTS 5, Cycle 4 database contains a set of 50 replicate weights to compute accurate standard errors for statistical testing procedures. These replicate weights were created using a jackknife minus one replication method; when analyzing one iteration or group of HINTS data, the proper denominator degrees of freedom (ddf) is 49. HINTS statistical analyses that involve more than one iteration of data—or a comparison for group differences in HINTS 5 Cycle 3– will typically utilize a set of 50*k replicate weights, where they can be viewed as being created using a stratified jackknife method with k as the number of strata or groups, and 49*k as the appropriate ddf. Analysts who were merging two iterations of data and making comparisons should adjust the ddf to be 98 (49*2), etc.

Taylor Series: The HINTS 5, Cycle 4 database contains two variables that can be used to calculate standard errors using the Taylor series, namely VAR_STRATUM and VAR_CLUSTER (see VAR_STRATUM and VAR_CLUSTER variables in the previous section for strata definitions.). The degrees of freedom for the Taylor series, 98, is based on 50 PSUs in each of the two sampling strata (#psus - #strata = 50*2 – 2 = 98).

Statistical Software Example Code

This section provides some coding examples using SAS, SPSS, and Stata for common types of statistical analyses using HINTS 5, Cycle 4 data. For SAS and Stata, you'll see two sets of code: one when using replicate methods for variance estimation, and one for Taylor Series linearization. For replicate methods, these examples will incorporate both the final sample weight (to get population-level point estimates) and the set of 50 jackknife replicate weights to get the proper standard error. For Taylor Series, the code will incorporate the final sample weight and the two variance codes to compute variance estimates. Although these examples specifically use HINTS 5, Cycle 4 data, the concepts used here are generally applicable to other types of analyses. We will consider an analysis that includes gender, education level (edu as a new variable) and two questions that are specific to the HINTS data: seekcancerinfo & generalhealth.

Analyzing Data Using SAS

Prior to using the HINTS 5, Cycle 4 SAS data, it is important to apply the SAS formats. To do this, follow the steps below.

1. Download all SAS HINTS 5 Cycle 4 data and documents to a folder on your computer. This should be the same folder where you create the SAS library in step #2.

2. Using SAS, create a permanent library to point to the folder where your data has been downloaded to (be sure to select, "enable at startup").

3. Open the SAS program *"HINTS 5 Cycle 4 Formats.sas."*

4. Change the file location specification in the "library" statement at the top of the program to the location where you want the format library to be stored before you run this program.

5. Run the program *"HINTS 5 Cycle 4 Formats.sas"* to create a permanent SAS format library that is used to analyze the HINTS dataset.

6. Open the SAS program "*HINTS 5 Cycle 4 Format Assignments.sas*."

7. Change the file location specification in the OPTIONS statement at the top of the program to the name of the library where you placed the formats. Also insert the library name for the SET and DATA statements and assign a name to the formatted data in the DATA statement.

8. Run the program "*HINTS 5 Cycle 4 Format Assignments.sas"* to create the formatted SAS data set.

Note:

1. Make sure to run the program *"HINTS 5 Cycle 4 Formats.sas"* BEFORE you run "*HINTS 5 Cycle 4 Format Assignments.sas*" to create the formatted HINTS dataset.
2. If you are getting an error statement saying that SAS is unable to find the formats, make sure you have run the OPTIONS statement that includes the correct library name where the formats can be found.

This section gives some SAS (Version 9.3 and higher) coding examples for common types of statistical analyses using HINTS 5, Cycle 4 data. Subsection 1 shows how to complete common analyses using replicate weights, and subsection 2 shows analyses using the Taylor series linearization approach. For either approach, we begin by doing data management of the HINTS 5 Cycle 4 data in a SAS DATA step. We first decided to exclude all "Missing data (Not Ascertained)" and "Multiple responses selected in error"

responses from the analyses. By setting these values to missing (.), SAS will exclude these responses from procedures where these variables are specifically accessed. When recoding existing variables, it is generally recommended to create new variables, rather than over-writing the existing variables. Note: New variables should always be compared to original source variables in a SAS PROC FREQ procedure to verify proper coding.

SAS Data Management Code: Recoding Variables and Creating and Applying New Formats

```
*This is used to call up the formats, substitute your library name
in the parentheses;
options fmtsearch=(hints5c4);

proc format;     *First create some temporary formats;
      Value Genderf
      1 = "Male"
      2 = "Female";

      Value Educationf
      1 = "Less than high school"
      2 = "12 years or completed high school"
      3 = "Some college"
      4 = "College graduate or higher";

      value seekcancerinfof
      1 = "Yes"
      0 = "No";

      Value Generalf
      1 = "Excellent"
      2 = "Very good"
      3 = "Good"
      4 = "Fair"
      5 = "Poor";
run;

data hints5cycle4;
      set hints5c4.hints5_cycle4_public;

      /*Recode negative values to missing*/
      **Note that the gender variable name has changed compared to
      previous cycles**;
      if birthgender = 1 then gender = 1;
      if birthgender = 2 then gender = 2;
      if birthgender in (-9) then gender = .;

      /*Recode education into four levels, and negative values to
      missing*/
      if education in (1, 2) then edu = 1;
      if education = 3 then edu = 2;
      if education in (4, 5) then edu = 3;
      if education in (6, 7) then edu = 4;
```

(code continued on the next page)

```
      if education in (-9) then edu = .;
      /*Recode seekcancerinfo to 0- 1 format for proc rlogist
      procedure, and negative values to missing */
      if seekcancerinfo = 2 then seekcancerinfo = 0;
      if seekcancerinfo in (-9, -6, -2, -1) then seekcancerinfo = .;

      /*Recode negative values to missing for proc regress procedure*/
      if generalhealth in (-5, -9) then generalhealth = .;

      /*Apply formats to recoded variables */
      format gender genderf. edu educationf. seekcancerinfo
      seekcancerinfof. generalhealth generalf.;
run;
```

SAS Replicate Weights Variance Estimation Method

Frequency Table and Chi-Square Test

We are now ready to begin using SAS 9.3 to examine the relationships among these variables. Using **PROC SURVEYFREQ**, we will first generate a cross-frequency table of education by gender, along with a (Wald) Chi-squared test of independence. Note the syntax of the overall sample weight, PERSON_FINWT0, and those of the jackknife replicate weights, PERSON_FINWT1—PERSON_FINWT50. The jackknife adjustment factor for each replicate weight is 0.98. This syntax is consistent for all procedures. Other datasets that incorporate replicate weight jackknife designs will follow a similar syntax.

```
proc surveyfreq data = hints5cycle4 varmethod = jackknife;
      weight PERSON_FINWT0;
      repweights PERSON_FINWT1-PERSON_FINWT50 / df = 49 jkcoefs = 0.98;
      tables edu*gender / row col wchisq;
run;
```

The *tables* statement defines the frequencies that should be generated. Standalone variables listed here result in one-way frequencies, while a "*" between variables will define cross-frequencies. The *row* option produces row percentages and standard errors, allowing us to view stratified percentages. Similarly, the *col* option produces column percentages and standard errors, allowing us to view stratified percentages. The option *wchisq* requests Wald chi-square test for independence. Other tests and statistics are also available; see the SAS 9.3 Product Documentation Site for more information.

For the purposes of computing appropriate degrees of freedom for the estimator of the HINTS5-Cycle 4 differences, we can assume, as an approximation, that the sample is a simple random sample of size 50 (corresponding to the 50 replicates: each replicate provides a "pseudo sample unit") from a normal distribution. The denominator degrees of freedom (df) is equal to 49*k, where k is the number of iterations of data used in this analysis.

Variance Estimation	
Method	Jackknife
Replicate Weights	hints5cycle4
Number of Replicates	50

Edu	gender	Frequency	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent	Column Percent	Std Err of Col Percent
Less than high school	Male	106	3.8988	0.6141	49.1842	4.4382	8.0321	1.2691
	Female	159	4.0282	0.4215	50.8158	4.4382	7.8279	0.8171
	Total	265	7.927	0.8001	100			
12 years or completed high school	Male	255	11.1273	0.8009	49.238	2.1605	22.9235	1.6142
	Female	436	11.4717	0.4703	50.762	2.1605	22.2928	0.9303
	Total	691	22.5989	0.8823	100			
Some college	Male	462	19.0926	0.8163	48.9482	1.2862	39.3331	1.6532
	Female	600	19.9131	0.4632	51.0518	1.2862	38.697	0.7915
	Total	1,062	39.0057	0.8849	100			
College graduate or higher	Male	698	14.4221	0.1151	47.3349	0.1949	29.7113	0.349
	Female	946	16.0462	0.1139	52.6651	0.1949	31.1823	0.2326
	Total	1,644	30.4683	0.1961	100			
Total	Male	1,521	48.5408	0.3126			100	
	Female	2,141	51.4592	0.3126			100	
	Total	3,662	100					

Frequency Missing =203

Wald Chi-Square Test	
Chi-Square	10.9249
F Value	3.6416
Num DF	3
Den DF	49
Pr > F	0.0189
Adj F Value	3.4930
Num DF	3
Den DF	47
Pr > Adj F	0.0227
Sample Size = 3,662	

The row percentages above show that a higher weighted proportion of college graduates in the sample are women (53%) than men (47%). Respondents with less than a high school diploma include more women (51%) than men (49%). The statistic for the Chi-square test of independence and its associated p-value indicate that the distributions of educational attainment between men and women are significantly different.

Logistic Regression

This example demonstrates a multivariable logistic regression model using **PROC SURVEYLOGISTIC**; recall that the response should be a dichotomous 0-1 variable.

```
/*Multivariable logistic regression of gender and education on
SeekCancerInfo*/
proc surveylogistic data= hints5cycle4 varmethod=jackknife;
      weight PERSON_FINWT0;
      repweights PERSON_FINWT1-PERSON_FINWT50 / df=49 jkcoefs=0.98;
      class edu (ref="Less than high school")
            gender (ref="Male")/param=REF;
      model seekcancerinfo (descending) = gender edu /tech=newton
      xconv=1e-8 CLPARM EXPB;
run;
```

The response variable should be on the left-hand side of the equal sign in the model statement, while all covariates should be listed on the right-hand side. The *descending* option requests the probability of seekcancerinfo= “Yes” to be modeled. The “Male” is the reference group for gender effect, while “Less than high school” is the reference group for education level effect. The option *tech=newton* requests the Newton-Raphson algorithm. The option *xconv=1e-8* helps to avoid early termination of the iteration.

Variance Estimation	
Method	Jackknife
Replicate Weights	hints5cycle4
Number of Replicates	50

Type 3 Analysis of Effects				
Effect	**F Value**	**Num DF**	**Den DF**	**Pf > F**
Gender	7.32	1	49	0.0094
Education	23.82	3	49	<.0001

Analysis of Maximum Likelihood Estimates							
Parameter	**DF**	**Estimate**	**Standard Error**	**T value**	**Pr > |t|**	**95% confidence limits**	
Intercept	49	-1.6466	0.3515	-4.68	<.0001	-2.353	-0.9402
Gender	49	0.3627	0.1341	2.71	0.0094	0.0933	0.6321
12 years or completed high school	49	0.8865	0.3516	2.52	0.015	0.1799	1.5931
Some College	49	1.2524	0.3337	3.75	0.0005	0.5819	1.9229
College graduate or higher	49	1.855	0.3387	5.48	<.0001	1.1743	2.5358

Odds Ratio Estimates

Effect	Point Estimate	95% Confidence Limits	
Female vs Male	1.437	1.098	1.882
12 years or completed high school vs Less than high school	2.427	1.197	4.919
Some college vs Less than high school	3.499	1.789	6.841
College graduate or higher vs Less than high school	6.392	3.236	12.626

To identify levels/variables that display a significant difference in response, the rule of thumb is to examine odds ratios where the confidence interval does not contain 1 (by default, SAS will use alpha=.05 to determine statistical significance; the user can change this value by specifying a different threshold in the code). However, significance may also be garnered from the test of whether the associated beta parameter is equal to 0 (see "Analysis of Maximum Likelihood Estimates" table above, "Estimate" column). According to this model, women appear to have 1.44 higher odds as men to have searched for cancer information.

Linear Regression

This example demonstrates a multivariable linear regression model using **PROC SURVEYREG**; recall that the response should be a continuous variable. For the purposes of this example, we decided to use an outcome with five levels as a continuous variable (GENERALHEALTH). Note that higher values on GENERALHEALTH indicate poorer self-reported health status.

```
/*Multivariable linear regression of gender and education on
GeneralHealth*/
proc surveyreg data= hints5cycle4 varmethod=jackknife;
      weight PERSON_FINWT0;
      repweights PERSON_FINWT1-PERSON_FINWT50 / df=49 jkcoefs=0.98;
      class edu (ref="Less than high school") gender (ref="Male");
      model generalhealth = edu gender /solution;
run;
```

Variance Estimation	
Method	Jackknife
Replicate Weights	hints5cycle4
Number of Replicates	50

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	2.77999	0.103779	26.79	<.0001
12 years or completed high school	0.00868	0.11897	0.07	0.9421
Some College	-0.1849	0.113105	-1.63	0.1085
College graduate or higher	-0.6182	0.093999	-6.58	<.0001
Female	0.03904	0.048801	0.8	0.4275

The table labeled Estimated Regression of Coefficients shows that respondents who were college graduates or higher reported better general health than those with less than a high school education ($p < 0.0001$) when controlling for all other variables in the model. Keep in mind that the outcome, general health, is coded such that lower scores correspond to better health. However, there's no significant difference in health score between males and females (p=0.4275), those with some college vs those without a high school diploma (p=.1085), and between those with only a high school diploma and those without a high school diploma (p=0.9421).

Tests of Model Effects

Contrast	Num DF	F Value	Pr > F
Model	4	36.19	<.0001
Intercept	1	7321.45	<.0001
Education	3	48.22	<.0001
Gender	1	0.64	0.4275

The table labeled Test of Model Effects also shows that the association between gender and general health is not significant, but the association between education and general health is significant.

SAS Taylor Series Linearization Variance Estimation Method

Frequency Table and Chi-Square Test

We are now ready to begin using SAS 9.3 to examine the relationships among these variables. Using **PROC SURVEYFREQ**, we will first generate a cross-frequency table of education by gender, along with a (Wald) Chi-squared test of independence. Note the syntax of the strata VAR_STRATUM, cluster VAR_CLUSTER, and overall sample weight PERSON_FINWT0. This syntax is consistent for all procedures. Other analyses that use Taylor Series approximation will follow a similar syntax.

```
proc surveyfreq data = hints5cycle4
     varmethod = TAYLOR;
     strata VAR_STRATUM;
     cluster VAR_CLUSTER;
     weight PERSON_FINWT0;
     tables edu*gender / row col wchisq;
run;
```

The *tables* statement defines the frequencies that should be generated. Standalone variables listed here result in one-way frequencies, while a "*" between variables will define cross-frequencies. The *row* option produces row percentages and standard errors, allowing us to view stratified percentages. Similarly, the *col* option produces column percentages and standard errors, allowing us to view stratified percentages. The option *wchisq* requests Wald chi-square test for independence. Other tests and statistics are also available; see the SAS 9.3 Product Documentation Site for more information.

edu	gender	Frequency	Percent	Std Err of Percent	Row Percent	St Err of Row Percent	Column Percent	Std Err of Col Percent
Less than High School	Male	106	3.8988	0.7069	49.1842	5.6339	8.0321	1.3919
	Female	159	4.0282	0.5395	50.8158	5.6339	7.8279	1.0112
	Total	265	7.927	0.8892	100			
12 years or Completed	Male	255	11.1273	1.1769	49.238	3.4918	22.9235	2.2294

edu	gender	Frequency	Percent	Std Err of Percent	Row Percent	St Err of Row Percent	Column Percent	Std Err of Col Percent
High School	Female	436	11.4717	0.7107	50.762	3.4918	22.2928	1.2213
	Total	691	22.5989	1.1475	100			
Some College	Male	462	19.0926	1.2703	48.9482	2.4827	39.3331	2.336
	Female	600	19.9131	1.1078	51.0518	2.4827	38.697	1.4458
	Total	1,062	39.0057	1.3956	100			
College Graduate or Higher	Male	698	14.4221	0.9689	47.3349	2.0201	29.7113	1.9691
	Female	946	16.0462	0.6855	52.6651	2.0201	31.1823	1.378
	Total	1,644	30.4683	1.1643	100			
Total	Male	1,521	48.5408	1.5457			100	
	Female	2,141	51.4592	1.5457			100	
	Total	3,662	100					

Frequency Missing = 203

Wald Chi-Square Test	
Chi-Square	0.3685
F Value	0.1228
Num DF	3
Den DF	98
Pr > F	0.9464
Adj F Value	0.1203
Num DF	3
Den DF	96
Pr > Adj F	0.9479
Sample Size = 3,662	

The row percentages above show that a higher weighted proportion of college graduates in the sample are women (53%) than men (47%). Respondents with less than a high school diploma include more women (51%) than men (49%). The Chi-squared test of independence statistic and associated p value suggest that one should accept the null hypothesis that the two variables are not associated, which indicates that there is not a significant difference between the distributions of educational attainment for these two groups.

Logistic Regression

This example demonstrates a multivariable logistic regression model using **PROC SURVEYLOGISTIC**; recall that the response should be a dichotomous 0-1 variable.

```
/*Multivariable logistic regression of gender and education on
SeekCancerInfo*/
proc surveylogistic data= hints5cycle4 varmethod=TAYLOR;
      strata VAR_STRATUM;
      cluster VAR_CLUSTER;
      weight PERSON_FINWT0;
      class edu (ref="Less than high school")
            gender (ref="Male")/param=REF;
      model seekcancerinfo (descending) = gender edu /tech=newton
      xconv=1e-8 CLPARM EXPB;
run;
```

The response variable should be on the left-hand side (LHS) of the equal sign in the model statement, while all covariates should be listed on the right-hand side (RHS). The *descending* option requests the probability of seekcancerinfo="Yes" to be modeled. The "Male" is the reference group for gender effect, while "Less than high school" is the reference group for education level effect. The option *tech=newton* requests the Newton-Raphson algorithm. The option *xconv=1e-8* helps to avoid early termination of the iteration.

Variance Estimation	
Methods	Taylor Series
Variance Adjustment	Degrees of Freedom (DF)

Type 3 Analysis of Effects

Effect	F Value	Num DF	Den DF	Pr > F
Gender	7.25	1	98	0.0083
Education	24.79	3	96	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	t Value	Pf > |t|	95% Confidence Limits	
Intercept	98	-1.6466	0.3035	-5.43	<.0001	-2.2489	-1.0443
Gender (Female)	98	0.3627	0.1347	2.69	0.0083	0.0954	0.6299
12 years or completed high school	98	0.8865	0.3023	2.93	0.0042	0.2866	1.4863
Some college	98	1.2524	0.2896	4.32	<.0001	0.6776	1.8271
College graduate or higher	98	1.855	0.2906	6.38	<.0001	1.2783	2.4318

Odds Ratio Estimates

Effect	Point Estimate	95% Confidence Limits	
Female vs Male	1.437	1.1	1.877
12 years or completed high school vs Less than high school	2.427	1.332	4.421
Some college vs Less than high school	3.499	1.969	6.216
College graduate or higher vs Less than high school	6.392	3.591	11.379

To identify levels/variables that display a significant difference in response, the rule of thumb is to examine odds ratios where the confidence interval does not contain 1 (by default, SAS will use alpha=.05 to determine statistical significance; the user can change this value by specifying a different threshold in the code). However, significance may also be garnered from the test of whether the associated beta parameter is equal to 0 (see "Analysis of Maximum Likelihood Estimates" table above). According to this model, women appear to be statistically more likely than men to have searched for cancer information.

Linear Regression

This example demonstrates a multivariable linear regression model using **PROC SURVEYREG**; recall that the response should be a continuous variable. For the purposes of this example, we decided to use an outcome with five levels as a continuous variable (GENERALHEALTH). Note that higher values on GENERALHEALTH indicate poorer self-reported health status.

```
/*Multivariable linear regression of gender and education on
GeneralHealth*/
proc surveyreg data= hints5cycle4 varmethod=TAYLOR;
      strata VAR_STRATUM;
      cluster VAR_CLUSTER;
      weight PERSON_FINWT0;
      class edu (ref="Less than high school") gender (ref="Male");
      model generalhealth = edu gender/solution;
run;
```

	Variance Estimation	
Method	Taylor Series	
Variance Adjustment	Degrees of Freedom (DF)	

Estimated Regression of Coefficients

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	2.77999	0.10852	25.62	<.0001
Female	0.03904	0.04536	0.86	0.3915
12 years or completed high school	0.00868	0.11491	0.08	0.9399
Some college	-0.1849	0.11319	-1.63	0.1056
College graduate or higher	-0.6182	0.1013	-6.1	<.0001

Compared to those respondents with less than a high school education, those who were a college graduate or higher on average reported significantly better general health (i.e., the negative beta coefficient indicates that the average health score is lower among those who were a college graduate or higher, and the health variable is coded such that lower scores correspond to better health), controlling for all variables in the model. When comparing to those with less than a high school education, those with some college or a high school education do not report statistically different general health ($p > .05$). We do not interpret the estimates for Female or for High School because the corresponding p-value is greater than .05.

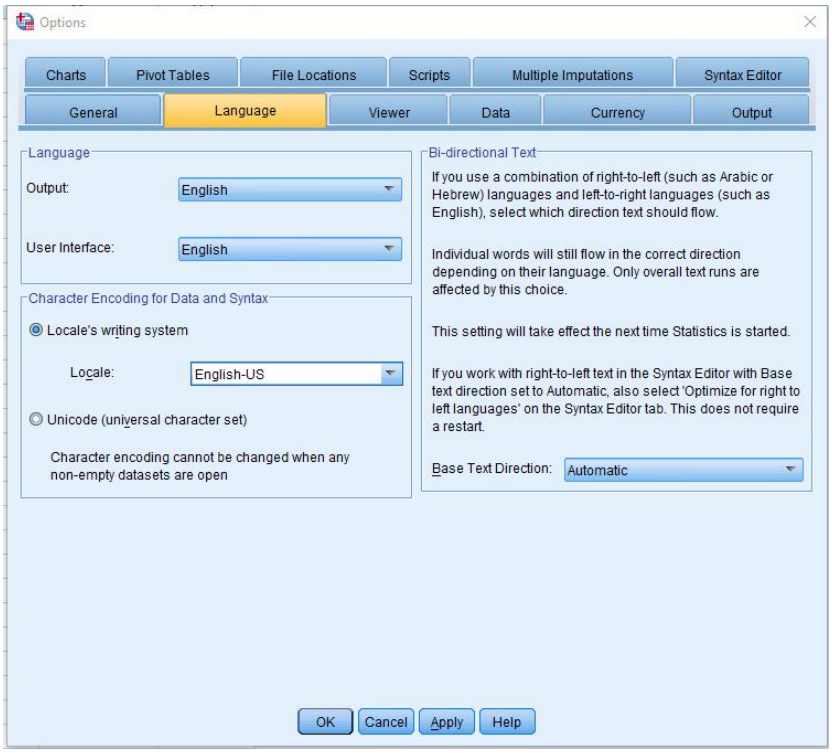
Test of Model Effects

Contrast	Num DF	F Value	Pr > F
Overall Model	4	33.54	<.0001
Intercept	1	6317.61	<.0001
Gender	1	0.74	0.3915
Education	3	44.46	<.0001

From the above table, we can see that gender is **not** significantly associated with general health, but education is significantly associated with general health, adjusting for all variables in the model.

Analyzing Data Using SPSS—Taylor Series

Prior to opening the HINTS 5, Cycle 4 SPSS data, it is important to ensure that your SPSS environment is set up to be compatible with the dataset. Specifically, the language encoding (i.e., the way that character data are stored and accessed) must match between your environment and the dataset. We recommend locale encoding in U.S. English over Unicode encoding. To ensure compatibility, you must update the language encoding manually through the graphic user interface (GUI). In a new SPSS session, from the empty dataset window, select "Edit" > "Options…" from the menu bar. In the pop-up box, select the "Language" tab. In this tab, look for the "Character Encoding for Data and Syntax" section. Select the "Locale's writing system" option and English-US or en-US from the "Locale:" dropdown list. "English-US" and "en-US" from the drop down are the common aliases used by SPSS to describe U.S. English encoding; if you do not see these specific aliases verbatim, choose the English alias that is most similar. Click "OK" to save your changes. You may now open the HINTS SPSS data without compatibility issues.



This section gives some SPSS (Version 25 and higher) coding examples for common types of statistical analyses using HINTS 5, Cycle 4 data. We begin by creating an analysis plan using the Complex Samples analysis procedures to specify the sample design; PERSON_FINWT0 is the sample weight

variable (the final weight for the composite sample, no group differences found), VAR_STRATUM is the stratum variable, and VAR_CLUSTER is the cluster variable. The subcommand SRSESTIMATOR specifies the variance estimator under the simple random sampling assumption. The default value is WR (with replacement), and it includes the finite population correction in the variance computation. The subcommand PRINT is used to control output from CSPLAN, and the syntax PLAN means to display a summary of plan specifications. The subcommand DESIGN with keyword STRATA identifies the sampling stratification variable, and the keyword cluster CLUSTER identifies the grouping of sampling units for variance estimation. The subcommand ESTIMATOR specifies the variance estimation method used in the analysis. The syntax TYPE=WR requires the estimation method of selection with replacement.

```
* Analysis Preparation Wizard.
*Substitute your file path and plan file name in the parentheses of /PLAN FILE=.
CSPLAN ANALYSIS
 /PLAN FILE='(sample.csaplan)'
 /PLANVARS ANALYSISWEIGHT=PERSON_FINWT0
 /SRSESTIMATOR TYPE=WOR
 /PRINT PLAN
 /DESIGN STRATA=VAR_STRATUM CLUSTER=VAR_CLUSTER
 /ESTIMATOR TYPE=WR.
```

We completed data management of the HINTS 5 Cycle 4 data in a SPSS RECODE step. We first decided to exclude all “Missing data (Not Ascertained)” and “Multiple responses selected in error” responses from the analyses. By setting these values to missing (SYSMIS), SPSS will exclude these responses from procedures where these variables are specifically accessed. For logistic regression modeling in the CSLOGISTIC procedure, SPSS by default always uses the last (highest) level of category of the covariates as the reference, similar to SAS. Users in SPSS cannot define the reference category by themselves unless they reorder the categories to create the desired value as the reference, such as using reverse coding (see example below). To make SPSS results comparable with SAS, we reverse coded the variables in SPSS. When recoding existing variables, it is generally recommended to create new variables, rather than over-writing the existing variables. Note: New variables should always be compared to original source variables in a SPSS CROSSTABS procedure to verify proper coding.

```
*Recode negative values to missing.
DATASET ACTIVATE DataSet1.
RECODE BirthGender (1=1) (2=2) (ELSE=SYSMIS) INTO gender.
VARIABLE LABELS gender 'gender'.
EXECUTE.

*Recode education into four levels, and negative values to missing.
RECODE Education (3=2) (1 thru 2=1) (4 thru 5=3) (6 thru 7=4) (ELSE=SYSMIS) INTO edu.
VARIABLE LABELS edu 'edu'.
EXECUTE.

*Recode seekcancerinfo to 0- 1 format for CSLOGISTIC procedure, and negative values to missing.
RECODE SeekCancerInfo (2=0) (1=1) (ELSE=SYSMIS) INTO seekcancerinfo_recode.
VARIABLE LABELS seekcancerinfo_recode 'seekcancerinfo_recode'.
EXECUTE.

*Recode negative values to missing for CSGLM procedure.
RECODE GeneralHealth (1 thru 5=Copy) (ELSE=SYSMIS) INTO genhealth_recode.
VARIABLE LABELS genhealth_recode 'genhealth_recode'.
EXECUTE.
```

(code continued on the next page)

```
*Reverse coding.
RECODE gender (1=2) (2=1) (ELSE=Copy) INTO flippedgender.
VARIABLE LABELS flippedgender 'flippedgender'.
EXECUTE.

*Reverse coding.
RECODE edu (1=4) (2=3) (3=2) (4=1) (ELSE=Copy) INTO flippededu.
VARIABLE LABELS flippededu 'flippededu'.
EXECUTE.

*Add value labels to recoded variables.
VALUE LABELS gender 1 "Male" 2 "Female".
VALUE LABELS flippedgender 2 "Male" 1 "Female".
VALUE LABELS edu 1 "Less than high school" 2 "12 years or completed high school" 3 "Some college"
4 "College graduate or higher".
VALUE LABELS flippededu 4 "Less than high school" 3 "12 years or completed high school" 2
"Some college" 1 "College graduate or higher".
VALUE LABELS seekcancerinfo_recode 1 "Yes" 0 "No".
VALUE LABELS genhealth_recode 1 "Excellent" 2 "Very good" 3 "Good" 4 "Fair" 5 "Poor".
```

Frequency Table and Chi-Square Test

We are now ready to begin using SPSS v25 to examine the relationships among these variables. Using **CSTABULATE**, we will first generate a cross-frequency table of education by gender. Note that we specify the file that contains the sample design specification using the subcommand PLAN. This syntax is consistent for all procedures. Other analyses using the same sample design will follow a similar syntax.

```
* Complex Samples Crosstabs.
CSTABULATE
/PLAN FILE="(plan filename)"
/TABLES VARIABLES=edu BY gender
/CELLS POPSIZE ROWPCT COLPCT TABLEPCT
/STATISTICS SE COUNT
/TEST INDEPENDENCE
/MISSING SCOPE=TABLE CLASSMISSING=EXCLUDE.
```

The TABLES subcommand defines the tabulation variables, where the syntax "BY" indicates the two-way crosstabulation. The CELLS subcommand specifies the summary value estimates to be displayed in the table. The *POPSIZE* option produces population size estimates for each cell and marginal. The *ROWPCT* option produces row percentages and standard errors. Similarly, the *COLPCT* option produces column percentages and standard errors. The *TABLEPCT* option produces table percentages and standard errors for each cell. The STATISTICS subcommand specifies the statistics to be displayed with the summary value estimates. The *SE* option produces the standard error for each summary value, and the *COUNT* option produces unweighted counts. The TEST subcommand specifies tests for the table. The *INDEPENDENCE* option produces the test of independence for the two-way crosstabulations. The MISSING subcommand specifies how missing values are handled. The *SCOPE* statement specifies which cases are used in the analyses. The *TABLE* option specifies that cases with all valid data for the tabulation variables are used in the analyses. The *CLASSMISSING* statement specifies whether user-defined missing values are included or excluded. The *EXCLUDE* option specifies user-defined missing values to be excluded in the analysis.

Crosstabs Edu*Gender

Edu			Gender		
			Male	Female	Total
Less than high school	Population Size	Estimate	9496143.974	9811170.229	19307314.204
		Standard Error	1820689.445	1347327.087	2361785.877
		Unweighted Count	106	159	265
	% within edu	Estimate	49.2%	50.8%	100.0%
		Standard Error	5.6%	5.6%	0.0%
		Unweighted Count	106	159	265
	% within gender	Estimate	8.0%	7.8%	7.9%
		Standard Error	1.4%	1.0%	0.9%
		Unweighted Count	106	159	265
	% of Total	Estimate	3.9%	4.0%	7.9%
		Standard Error	0.7%	0.5%	0.9%
		Unweighted Count	106	159	265
12 years or completed high school	Population Size	Estimate	27101924.979	27940785.805	55042710.785
		Standard Error	3090503.825	1834353.410	3353705.737
		Unweighted Count	255	436	691
	% within edu	Estimate	49.2%	50.8%	100.0%
		Standard Error	3.5%	3.5%	0.0%
		Unweighted Count	255	436	691
	% within gender	Estimate	22.9%	22.3%	22.6%
		Standard Error	2.2%	1.2%	1.1%
		Unweighted Count	255	436	691
	% of Total	Estimate	11.1%	11.5%	22.6%
		Standard Error	1.2%	0.7%	1.1%
		Unweighted Count	255	436	691
Some college	Population Size	Estimate	46502569.296	48501067.307	95003636.603
		Standard Error	3401360.032	2964627.127	4311413.771
		Unweighted Count	462	600	1062
	% within edu	Estimate	48.9%	51.1%	100.0%
		Standard Error	2.5%	2.5%	0.0%
		Unweighted Count	462	600	1062
	% within gender	Estimate	39.3%	38.7%	39.0%
		Standard Error	2.3%	1.4%	1.4%
		Unweighted Count	462	600	1062
	% of Total	Estimate	19.1%	19.9%	39.0%
		Standard Error	1.3%	1.1%	1.4%
		Unweighted Count	462	600	1062

College graduate or higher	Population Size	Estimate	35126967.297	39082541.229	74209508.526
		Standard Error	2447941.611	1717982.126	3041220.867
		Unweighted Count	698	946	1644
	% within edu	Estimate	47.3%	52.7%	100.0%
		Standard Error	2.0%	2.0%	0.0%
		Unweighted Count	698	946	1644
	% within gender	Estimate	29.7%	31.2%	30.5%
		Standard Error	2.0%	1.4%	1.2%
		Unweighted Count	698	946	1644
	% of Total	Estimate	14.4%	16.0%	30.5%
		Standard Error	1.0%	0.7%	1.2%
		Unweighted Count	698	946	1644
Total	Population Size	Estimate	118227605.546	125335564.571	243563170.117
		Standard Error	5555587.936	4623884.337	6956299.695
		Unweighted Count	1521	2141	3662
	% within edu	Estimate	48.5%	51.5%	100.0%
		Standard Error	1.5%	1.5%	0.0%
		Unweighted Count	1521	2141	3662
	% within gender	Estimate	100.0%	100.0%	100.0%
		Standard Error	0.0%	0.0%	0.0%
		Unweighted Count	1521	2141	3662
	% of Total	Estimate	48.5%	51.5%	100.0%
		Standard Error	1.5%	1.5%	0.0%
		Unweighted Count	1521	2141	3662

The row percentages above show that a higher weighted proportion of college graduates in the sample are women (53%) than men (47%). Respondents with less than a high school diploma include more women (51%) than men (49%).

Tests of Independence

		Chi-Square	Adjusted F	df1	df2	Sig.
edu * gender	Pearson	.954	.100	2.852	279.506	.955
	Likelihood Ratio	.954	.100	2.852	279.506	.955

Pearson chi-square test statistic and Likelihood Ratio test statistic and their associated p-values suggest that one should accept the null hypothesis that the two variables are not associated, which indicates that there is not a significant difference between the distributions of educational attainment for men and women.

The results of these tests conducted in SPSS based on Taylor Series linearization contradict the results conducted in SAS using replication shown in the “Analyzing Data Using SAS” section. (In SAS, the distributions of educational attainment between men and women were determined to be statistically different when analyzing the results using replicate weights). This is a good example of how the variance

estimation method used can affect the outcome of a statistical test. Both education and gender are variables used in the raking process as part of the HINTS weighting procedure. As a result, the standard errors based on replication are much smaller than those based on Taylor Series linearization, which in turn results in significant differences in SAS but not in SPSS.

Note that the CSTABULATE procedure provides results for the Pearson Chi-square and Likelihood Ratio tests, but not for the Wald Chi-square test of independence. To get the results for the Wald Chi-square test of independence, users can conduct a logistic regression model in the CSLOGISTIC procedure in which the type of Chi-square test can be specified.

Logistic Regression

This example demonstrates a multivariable logistic regression model using **CSLOGISTIC**; recall that the response should be a categorical variable.

```
*Multivariable logistic regression of gender and education on SeekCancerInfo.
CSLOGISTIC  seekcancerinfo_recode (LOW) BY flippedgender flippededu
 /PLAN FILE='(sample.csaplan)'
 /MODEL flippedgender flippededu
 /CUSTOM  Label = 'Overall model minus intercept'
  LMATRIX = flippedgender 1/2 -1/2;
          flippededu 1/3 1/3 1/3 -1;
         flippededu 1/3 1/3 -1 1/3;
         flippededu 1/3 -1 1/3 1/3;
         flippededu -1 1/3 1/3 1/3
  /CUSTOM  Label = 'Gender'
 LMATRIX =  flippedgender 1/2 -1/2
  /CUSTOM  Label = 'Education overall'
  LMATRIX = flippededu 1/3 1/3 1/3 -1;
         flippededu 1/3 1/3 -1 1/3;
         flippededu 1/3 -1 1/3 1/3;
         flippededu -1 1/3 1/3 1/3
 /INTERCEPT INCLUDE=YES SHOW=YES
 /STATISTICS PARAMETER SE CINTERVAL TTEST EXP
 /TEST TYPE=CHISQUARE PADJUST=LSD
 /ODDSRATIOS FACTOR=[flippedgender(HIGH)]
 /ODDSRATIOS FACTOR=[flippededu(HIGH)]
 /MISSING CLASSMISSING=EXCLUDE
 /CRITERIA MXITER=100 MXSTEP=50 PCONVERGE=[1e-008 RELATIVE] LCONVERGE=[0]
CHKSEP=20 CILEVEL=95
 /PRINT SUMMARY COVB CORB VARIABLEINFO SAMPLEINFO.
```

The response variable should be on the left-hand side of the BY statement, while all covariates should be listed on the right-hand side. The (LOW) option indicates that the lowest category is the reference category, thus requests the probability of seekcancerinfo="Yes" to be modeled. The "Male" is the reference group for gender effect, while "Less than high school" is the reference group for education level effect. The subcommand MODEL specifies all variables in the model. The CUSTOM subcommand allows users to define custom hypothesis tests. The LMATRIX statement specifies coefficients of contrasts, which are used for studying the effects in the model. The INTERCEPT subcommand specifies whether to include or show the intercept in the final estimates. The STATISTICS subcommand specifies the statistics to be estimated and shown in the final result, where the syntax PARAMETER indicates the coefficient estimates, EXP indicates the exponentiated coefficient estimates, SE indicates the standard error for each coefficient estimate, CINTERVAL indicates the confidence interval for each coefficient estimate. The TEST subcommand specifies the type of test statistic and the method of adjusting the significance level to

be used for hypothesis tests that are requested on the MODEL and CUSTOM subcommands, where the syntax CHISQUARE indicates the Wald chi-square test, and LSD indicates the least significant difference. The ODDSRATIOS subcommand estimates odds ratios for certain factors. The subcommand MISSING specifies how to handle missing data. The subcommand CRITERIA offers controls on the iterative algorithm that is used for estimations. The option PCONVERGE= [1e-008 RELATIVE] helps to avoid early termination of the iteration. The subcommand PRINT is used to display optional output.

Sample Design Information

		N
Unweighted Cases	**Valid**	3658
	Invalid	207
	Total	3865
Population Size		243374915.695
Stage 1	**Strata**	2
	Units	100
Sampling Design Degrees of Freedom		98

Parameter Estimates

seekcancerinfo_recode		B	Std. Error	95% Confidence Interval		Hypothesis Test			Exp(B)	95% Confidence Interval for Exp(B)	
				Lower	Upper	t	df	Sig.		Lower	Upper
Yes	**(Intercept)**	-1.647	.303	-2.249	-1.045	-5.428	98.000	.000	.193	.106	.352
	Female	.363	.135	.096	.630	2.695	98.000	.008	1.437	1.100	1.877
	College Graduate or Higher	1.855	.290	1.279	2.431	6.386	98.000	.000	6.392	3.592	11.376
	Some College	1.252	.289	.678	1.827	4.326	98.000	.000	3.499	1.970	6.214
	12 Years or Completed High School	.886	.302	.287	1.486	2.934	98.000	.004	2.427	1.332	4.419

Odds Ratios

		seekcancerinfo_recode	Odds Ratio	95% Confidence Interval	
				Lower	Upper
Flipped Gender	**Female vs. Male**	Yes	1.437	1.100	1.877
Flipped Edu	**College graduate or higher vs. Less than high school**	Yes	6.392	3.592	11.376
	Some college vs. Less than high school	Yes	3.499	1.970	6.214
	12 years or completed high school vs. Less than high school	Yes	2.427	1.332	4.419

Overall Model Minus Intercept

df	Wald Chi-Square	Sig.
4.000	83.527	0.000

Gender

df	Wald Chi-Square	Sig.
1.000	7.261	0.007

Education Overall

df	Wald Chi-Square	Sig.
3.000	75.991	0.000

To identify levels/variables that display a significant difference in response, the rule of thumb is to examine odds ratios where the confidence interval does not contain 1 (by default, SPSS will use alpha=.05 to determine statistical significance; this value can be changed by the user using code). However, significance may also be garnered from the test of whether the associated beta parameter is equal to 0 (see "Parameter Estimates" table above). According to this model, women appear to be statistically more likely than men to have searched for cancer information.

Note that in SPSS we cannot get the overall model effect, even if we used the CUSTOM subcommand to conduct custom hypothesis tests.

Linear Regression

This example demonstrates a multivariable linear regression model using **CSGLM**; recall that the response should be a continuous variable. For the purposes of this example, we decided to use an outcome with five levels as a continuous variable (GENERALHEALTH). Note that higher values on GENERALHEALTH indicate poorer self-reported health status.

```
* Multivariable linear regression of gender and education on GeneralHealth.
CSGLM genhealth_recode BY flippedgender flippededu
 /PLAN FILE='(sample.csaplan)'
 /MODEL flippededu flippedgender
 /CUSTOM  Label = 'Overall model minus intercept'
  LMATRIX = flippedgender 1/2 -1/2;
          flippededu 1/3 1/3 1/3 -1;
          flippededu 1/3 1/3 -1 1/3;
          flippededu 1/3 -1 1/3 1/3;
          flippededu -1 1/3 1/3 1/3
  /CUSTOM  Label = 'Gender'
 LMATRIX =  flippedgender 1/2 -1/2
  /CUSTOM  Label = 'Education overall'
  LMATRIX =  flippededu 1/3 1/3 1/3 -1;
            flippededu 1/3 1/3 -1 1/3;
            flippededu 1/3 -1 1/3 1/3;
            flippededu -1 1/3 1/3 1/3
```

(code continued on the next page)

/INTERCEPT INCLUDE=YES SHOW=YES
/STATISTICS PARAMETER SE CINTERVAL TTEST

/PRINT SUMMARY VARIABLEINFO SAMPLEINFO
/TEST TYPE=F PADJUST=LSD
/MISSING CLASSMISSING=EXCLUDE
/CRITERIA CILEVEL=95.

Sample Design Information

		N
Unweighted Cases	**Valid**	3642
	Invalid	223
	Total	3865
Population Size		242654073.999
Stage 1	**Strata**	2
	Units	100
Sampling Design Degrees of Freedom		98

Parameter Estimates

Parameter	Estimate	Std. Error	95% Confidence Interval		Hypothesis Test		
			Lower	Upper	t	df	Sig.
(Intercept)	2.780	.108	2.565	2.995	25.631	98.000	.000
College Graduate or Higher	-.618	.101	-.819	-.417	-6.106	98.000	.000
Some College	-.185	.113	-.409	.040	-1.634	98.000	.105
12 Years or Completed High School	.009	.115	-.219	.237	.076	98.000	.940
Female	.039	.045	-.051	.129	.861	98.000	.391

Compared to those respondents with less than a high school education, those who were a college graduate or higher on average reported significantly better general health (i.e., the negative beta coefficient indicates that the average health score is lower among those with some college, and the health variable is coded such that lower scores correspond to better health), controlling for all variables in the model. We do not interpret the estimates for those who had some college, those who completed high school, or for female because the corresponding p-values are greater than .05.

Overall Model Minus Intercept

df1	df2	Wald F	Sig.
4.000	95	32.553	0.000

Gender

df1	df2	Wald F	Sig.
1.000	98	.742	.391

Education Overall

df1	df2	Wald F	Sig.
3.000	96	43.604	0.000

From the above table, we can see that education, but not gender, is significantly associated with general health.

Analyzing Data Using Stata

This section gives some Stata (Version 10.0 and higher) coding examples for common types of statistical analyses using HINTS 5, Cycle 4 data. Subsection 1 shows how to complete common analyses using replicate weights, and subsection 2 shows analyses using the Taylor Series linearization approach. For either approach, we begin by doing data management of the HINTS 5 data. We first decided to exclude all "Missing data (Not Ascertained)", "Multiple responses selected in error", "Question answered in error (Commission Error)", and "Inapplicable, coded 2 in SeekHealthInfo" responses from the analyses. By setting these values to missing (.), Stata will exclude these responses from analysis commands where these variables are specifically accessed. For logistic regression modeling within the svy: logit command, Stata expects the response variable to be dichotomous with values (0, 1), so this variable will also be recoded at this point. When recoding existing variables, it is generally recommended to create new variables rather than over-writing the existing variables. Note: New variables should always be compared to original source variables in a Stata **tabulate** command to verify proper coding.

```
use "file path\hints5_cycle4_public.dta"
* Recode negative values to missing
recode birthgender (1=1 "Male") (2=2 "Female") (nonmissing=.),
generate(gender)

label variable gender "Gender"

* Recode education into four levels, and negative values to missing

recode education (1/2=1 "Less than high school") (3=2 "12 years or
completed high school") (4/5=3 "Some college") (6/7=4 "College graduate
or higher") (nonmissing=.), generate(edu)
label variable edu "Education"
* Recode seekcancerinfo to 0-1 format, and negative values to
missing for svy: logit

replace seekcancerinfo = 0 if seekcancerinfo == 2
```

(code continued on the next page)

```
replace seekcancerinfo = . if seekcancerinfo == -1 | seekcancerinfo == -
2 | seekcancerinfo == -6 | seekcancerinfo == -9


label define seekcancerinfo2 0 "No" 1 "Yes"

label values seekcancerinfo seekcancerinfo2


* Recode negative values to missing for svy: regress

replace generalhealth = . if generalhealth == -5 | generalhealth == -9
```

Stata Replicate Weights Variance Estimation Method

Declare survey design

Stata requires that the survey design be declared for the dataset globally before any analysis. The declared survey design will be applied to all future survey commands unless another survey design is declared. In this example and declared design we are using PERSON_FINWT0 and its associated replicate weights (PERSON_FINTWT1 through PERSON_FINWT50).

```
* Declare survey design for the data set

svyset [pw= person_finwt0], jkrw(person_finwt1-
person_finwt50,multiplier(0.98)) vce(jack) mse
```

Cross-tabulation

```
* cross-tabulation: to obtain standard errors for total, row, and column you
must separately request each under different tabulate statements
svy: tabulate edu gender, cell format(%8.5f) percent se wald noadjust
svy: tabulate edu gender, row format(%8.5f) percent se wald noadjust
svy: tabulate edu gender, column format(%8.5f) percent se wald noadjust
```

The svy: tabulate command defines the frequencies that should be generated. Single variables listed in svy: tabulate results in one-way frequencies, while two variables will define cross-frequencies. The options cell, column, row request total cell, column, and row frequencies, respectively. These options must be individually run. The option percent requests the frequencies and are displayed in percentages. The options wald and noadjust together request the unadjusted Wald test for independence. Stata recommends the default Pearson test for independence. Other tests and statistics are also available; see the Stata website for more information: http://www.stata.com.

(results on subsequent pages)

Jknife *: for cell counts

Number of strata = 1

Number of obs = 3,662
Population size = 243,563,170
Replications = 50
Design df = 49

Education	Gender Male	Female	Total
Less tha	3.89884	4.02818	7.92703
	(0.61413)	(0.42149)	(0.80011)
12 years	11.12727	11.47168	22.59895
	(0.80092)	(0.47031)	(0.88231)
Some col	19.09261	19.91314	39.00575
	(0.81628)	(0.46324)	(0.88495)
College	14.42212	16.04616	30.46828
	(0.11508)	(0.11391)	(0.19608)
Total	48.54084	51.45916	1.0e+02
	(0.31264)	(0.31263)	

Key: cell percentage
(jackknife standard error of cell percentage)

Wald (Pearson):
Unadjusted chi2(3) = 10.9249
Unadjusted F(3, 49) = 3.6416 P = 0.0189
Adjusted F(3, 47) = 3.4930 P = 0.0227

```
Jknife *: for cell counts

Jknife *: for rows

Number of strata   =         1                Number of obs      =         3,662
                                              Population size    =   243,563,170
                                              Replications       =            50
                                              Design df          =            49
```

Education	Gender Male	Female	Total
Less tha	49.18418	50.81582	1.0e+02
	(4.43818)	(4.43818)	
12 years	49.23799	50.76201	1.0e+02
	(2.16045)	(2.16045)	
Some col	48.94820	51.05180	1.0e+02
	(1.28620)	(1.28620)	
College	47.33486	52.66514	1.0e+02
	(0.19493)	(0.19493)	
Total	48.54084	51.45916	1.0e+02
	(0.31264)	(0.31263)	

Key: row percentage
(jackknife standard error of row percentage)

```
Wald (Pearson):
  Unadjusted    chi2(3)          =   10.9249
  Unadjusted    F(3, 49)         =    3.6416     P = 0.0189
  Adjusted      F(3, 47)         =    3.4930     P = 0.0227
```

```
Jknife *: for cell counts

Jknife *: for columns

Number of strata   =         1              Number of obs      =         3,662
                                            Population size    =   243,563,170
                                            Replications       =            50
                                            Design df          =            49
```

Education	Gender Male	Female	Total
Less tha	8.03209	7.82792	7.92703
	(1.26913)	(0.81709)	(0.80011)
12 years	22.92352	22.29278	22.59895
	(1.61421)	(0.93025)	(0.88231)
Some col	39.33309	38.69697	39.00575
	(1.65317)	(0.79146)	(0.88495)
College	29.71131	31.18232	30.46828
	(0.34895)	(0.23260)	(0.19608)
Total	1.0e+02	1.0e+02	1.0e+02

```
  Key:  column percentage
        (jackknife standard error of column percentage)

  Wald (Pearson):
    Unadjusted    chi2(3)          =    10.9249
    Unadjusted    F(3, 49)         =     3.6416     P = 0.0189
    Adjusted      F(3, 47)         =     3.4930     P = 0.0227
```

For the purposes of computing appropriate degrees of freedom for the estimator of the HINTS 5, Cycle 4 differences, we can assume as an approximation that the sample is a simple random sample of size 50 (corresponding to the 50 replicates: each replicate provides a “pseudo sample unit”) from a normal distribution. The denominator degrees of freedom (df) is equal to 49*k, where k is the number of iterations of data used in this analysis. Stata uses the number of replicates minus one as the denominator degrees of freedom and does not provide the option for the user to specify the denominator degrees of freedom.

Logistic Regression

This example demonstrates a multivariable logistic regression model using **svy: logit** (to get parameters) and **svy, or: logit** (to get odds ratios); recall that the response should be a dichotomous 0-1 variable.

```
*   Define reference group for categorical variables for both svy: logit
and svy: regress
char gender [omit] 1
char edu [omit] 1
*   Multivariable logistic regression of gender and education on
seekcancerinfo xi: svy: logit seekcancerinfo i.gender i.edu
test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4 _cons, nosvyadjust
test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
test _Igender_2, nosvyadjust
test _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
xi: svy, or: logit seekcancerinfo i.gender i.edu
```

The **char** command defines the categorical variable with the reference group. The “Male” is the reference group for gender effect, while the “Less than high school” is the reference group for education level effect. These definitions will be applied to future commands until another **char** command redefines the reference group. The xi command will create proper dummy variables for i.gender and i.edu variables in the analysis commands. The response variable should be the first variable in the **svy: logit** command and be followed by all covariates. The **test** command tests the hypotheses about estimated parameters.

(results on subsequent pages)

```
. xi: svy: logit seekcancerinfo i.gender i.edu
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running logit on estimation sample)

Jackknife replications (50)
----+--- 1 ---+--- 2 ---+--- 3 ---+--- 4 ---+--- 5
..................................................    50

Survey: Logistic regression

Number of strata   =         1                  Number of obs      =        3,658
                                                Population size    =  243,374,916
                                                Replications       =           50
                                                Design df          =           49
                                                F(   4,     46)    =        20.12
                                                Prob > F           =       0.0000
```

seekcancerinfo	Coef.	Jknife * Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	.3626786	.1340651	2.71	0.009	.0932646	.6320926
_Iedu_2	.8864881	.3516168	2.52	0.015	.1798877	1.593089
_Iedu_3	1.252377	.3336548	3.75	0.000	.5818722	1.922881
_Iedu_4	1.855047	.3387297	5.48	0.000	1.174344	2.53575
_cons	-1.646597	.3515239	-4.68	0.000	-2.353011	-.9401837

```
Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_2 = 0
 ( 3)  [seekcancerinfo]_Iedu_3 = 0
 ( 4)  [seekcancerinfo]_Iedu_4 = 0
 ( 5)  [seekcancerinfo]_cons = 0

       F(  5,    49) =   17.15
            Prob > F =    0.0000
```

```
Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_2 = 0
 ( 3)  [seekcancerinfo]_Iedu_3 = 0
 ( 4)  [seekcancerinfo]_Iedu_4 = 0

       F(  4,    49) =   21.44
            Prob > F =    0.0000


Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0

       F(  1,    49) =    7.32
            Prob > F =    0.0094


Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Iedu_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_3 = 0
 ( 3)  [seekcancerinfo]_Iedu_4 = 0

       F(  3,    49) =   23.82
            Prob > F =    0.0000
```

```
. xi: svy, or: logit seekcancerinfo i.gender i.edu
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running logit on estimation sample)

Jackknife replications (50)
----+--- 1 ---+--- 2 ---+--- 3 ---+--- 4 ---+--- 5
..................................................    50

Survey: Logistic regression

Number of strata   =         1              Number of obs      =        3,658
                                            Population size    =  243,374,916
                                            Replications       =           50
                                            Design df          =           49
                                            F(   4,     46)    =        20.12
                                            Prob > F           =       0.0000
```

seekcancerinfo	Odds Ratio	Jknife * Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	1.437174	.1926749	2.71	0.009	1.097752	1.881544
_Iedu_2	2.426593	.8532308	2.52	0.015	1.197083	4.918918
_Iedu_3	3.498648	1.167341	3.75	0.000	1.789385	6.840639
_Iedu_4	6.391998	2.16516	5.48	0.000	3.23602	12.62589
_cons	.1927045	.0677402	-4.68	0.000	.0950824	.3905561

```
Note: _cons estimates baseline odds.
```

To identify levels/variables that display a significant difference in response, the rule of thumb is to examine odds ratios where the confidence interval does not contain 1 (by default, Stata will use alpha=.05 to determine statistical significance; this value can be changed using code). However, significance may also be garnered from the test of whether the associated beta parameter is equal to 0 (see first regression table above). According to this model, women appear to be 1.44 times as likely as men to have searched for cancer information.

Linear Regression

This example demonstrates a multivariable linear regression model using **svy: regress**; recall that the response should be a continuous variable. For the purposes of this example, we decided to use an outcome with five levels as a continuous variable (generalhealth). Note that higher values on generalhealth indicate poorer self-reported health status.

```
*   Multivariable linear regression of gender and education on generalhealth

xi: svy: regress generalhealth i.gender i.edu

test Igender2 Iedu2 Iedu3 Iedu4 _cons, nosvyadjust
test Igender2 Iedu2 Iedu3 Iedu4, nosvyadjust
test Igender2, nosvyadjust
test Iedu2 Iedu3 Iedu4, nosvyadjust
```

```
. xi: svy: regress generalhealth i.gender i.edu
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running regress on estimation sample)

Jackknife replications (50)
----+--- 1 ---+--- 2 ---+--- 3 ---+--- 4 ---+--- 5
..................................................    50

Survey: Linear regression

Number of strata   =         1                  Number of obs     =        3,642
                                                 Population size   =  242,654,074
                                                 Replications      =           50
                                                 Design df         =           49
                                                 F(   4,     46)   =        33.97
                                                 Prob > F          =       0.0000
                                                 R-squared         =       0.0741
```

generalhea~h	Coef.	Jknife * Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	.0390436	.0488008	0.80	0.428	-.0590254	.1371126
_Iedu_2	.008683	.11897	0.07	0.942	-.2303961	.2477621
_Iedu_3	-.1848788	.1131047	-1.63	0.109	-.4121712	.0424136
_Iedu_4	-.6182037	.0939988	-6.58	0.000	-.8071013	-.4293061
_cons	2.779991	.1037786	26.79	0.000	2.57144	2.988542

```
Unadjusted Wald test

 ( 1)  _Igender_2 = 0
 ( 2)  _Iedu_2 = 0
 ( 3)  _Iedu_3 = 0
 ( 4)  _Iedu_4 = 0
 ( 5)  _cons = 0

       F(  5,    49) = 2466.76
            Prob > F =    0.0000
```

```
Unadjusted Wald test

 ( 1)  _Igender_2 = 0
 ( 2)  _Iedu_2 = 0
 ( 3)  _Iedu_3 = 0
 ( 4)  _Iedu_4 = 0

       F(  4,    49) =   36.19
            Prob > F =    0.0000

Unadjusted Wald test

 ( 1)  _Igender_2 = 0

       F(  1,    49) =    0.64
            Prob > F =    0.4275

Unadjusted Wald test

 ( 1)  _Iedu_2 = 0
 ( 2)  _Iedu_3 = 0
 ( 3)  _Iedu_4 = 0

       F(  3,    49) =   48.22
            Prob > F =    0.0000
```

From the above table, it can be seen that, compared to those respondents with less than a high school education, those with a college degree or higher have a significantly negative linear association with the outcome (i.e., better reported health), controlling for all variables in the model. We do not interpret the gender variable or those with a high school education or some college education because they are non-significant.

Stata Taylor Series Linearization Variance Estimation Method

Declare survey design

Stata requires that the survey design be declared for the dataset globally before any analysis. The declared survey design will be applied to all future survey commands unless another survey design is declared. In this example and declared design we are using PERSON_FINWT0.

```
* Declare survey design for the data set (Taylor series)
svyset var_cluster [pw=person_finwt0], strata(var_stratum)
```

Cross-tabulation

```
* cross-tabulation
svy: tabulate edu gender, cell format(%8.5f) percent se wald noadjust
svy: tabulate edu gender, row format(%8.5f) percent se wald noadjust
svy: tabulate edu gender, column format(%8.5f) percent se wald noadjust
```

The **svy: tabulate** command defines the frequencies that should be generated. Single variables listed in **svy: tabulate** results in one-way frequencies, while two variables will define cross-frequencies. The options cell, column, row request total cell, column, and row frequencies, respectively. These options must be individually run. The option percent requests the frequencies and are displayed in percentages. The options wald and noadjust together request the unadjusted Wald test for independence. Stata recommends the default Pearson test for independence. Other tests and statistics are also available; see the Stata website for more information: http://www.stata.com.

```
(running tabulate on estimation sample)

Number of strata   =         2                  Number of obs      =         3,662
Number of PSUs     =       100                  Population size    =   243,563,170
                                                Design df          =            98
```

Education	Gender Male	Female	Total
Less tha	3.89884	4.02818	7.92703
	(0.70687)	(0.53954)	(0.88923)
12 years	11.12727	11.47168	22.59895
	(1.17688)	(0.71073)	(1.14748)
Some col	19.09261	19.91314	39.00575
	(1.27034)	(1.10777)	(1.39561)
College	14.42212	16.04616	30.46828
	(0.96885)	(0.68548)	(1.16432)
Total	48.54084	51.45916	1.0e+02
	(1.54571)	(1.54571)	

```
  Key:  cell percentage
        (linearized standard error of cell percentage)

  Wald (Pearson):
    Unadjusted    chi2(3)          =    0.3685
    Unadjusted    F(3, 98)         =    0.1228     P = 0.9464
    Adjusted      F(3, 96)         =    0.1203     P = 0.9479
```

(running tabulate on estimation sample)

Number of strata	=	2	
Number of PSUs	=	100	
Number of obs	=	3,662	
Population size	=	243,563,170	
Design df	=	98	

Education	Gender Male	Female	Total
Less tha	49.18418	50.81582	1.0e+02
	(5.63387)	(5.63387)	
12 years	49.23799	50.76201	1.0e+02
	(3.49179)	(3.49179)	
Some col	48.94820	51.05180	1.0e+02
	(2.48272)	(2.48272)	
College	47.33486	52.66514	1.0e+02
	(2.02009)	(2.02009)	
Total	48.54084	51.45916	1.0e+02
	(1.54571)	(1.54571)	

Key: row percentage
(linearized standard error of row percentage)

Wald (Pearson):

Unadjusted	chi2(3)	=	0.3685	
Unadjusted	F(3, 98)	=	0.1228	P = 0.9464
Adjusted	F(3, 96)	=	0.1203	P = 0.9479

```
(running tabulate on estimation sample)

Number of strata   =         2                  Number of obs      =         3,662
Number of PSUs     =       100                  Population size    =   243,563,170
                                                Design df          =            98
```

Education	Gender Male	Female	Total
Less tha	8.03209	7.82792	7.92703
	(1.39187)	(1.01116)	(0.88923)
12 years	22.92352	22.29278	22.59895
	(2.22944)	(1.22133)	(1.14748)
Some col	39.33309	38.69697	39.00575
	(2.33596)	(1.44579)	(1.39561)
College	29.71131	31.18232	30.46828
	(1.96913)	(1.37800)	(1.16432)
Total	1.0e+02	1.0e+02	1.0e+02

```
  Key:  column percentage
        (linearized standard error of column percentage)

  Wald (Pearson):
    Unadjusted    chi2(3)          =    0.3685
    Unadjusted    F(3, 98)         =    0.1228     P = 0.9464
    Adjusted      F(3, 96)         =    0.1203     P = 0.9479
```

Logistic Regression

This example demonstrates a multivariable logistic regression model using **svy: logit** (to get parameters) and **svy, or: logit** (to get odds ratios); recall that the response should be a dichotomous 0-1 variable.

```
*   Define reference group for categorical variables for both svy: logit
and svy: regress
char gender [omit] 1
char edu [omit] 1
*   Multivariable logistic regression of gender and education on
seekcancerinfo
```

(code continued on the next page)

```
xi: svy: logit seekcancerinfo i.gender i.edu

test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4 _cons, nosvyadjust
test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
test _Igender_2, nosvyadjust

test _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
xi: svy, or: logit seekcancerinfo i.gender i.edu
```

The **char** command defines categorical variable with reference group. The “Male” is the reference group for gender effect, while the “Less than high school” is the reference group for education level effect. These definitions will be applied to future commands until another char command redefines the reference group. The xi command will create proper dummy variables for i.gender and i.edu variables in the analysis commands. The response variable should be the first variable in **svy: logit** command and be

```
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running logit on estimation sample)

Survey: Logistic regression

Number of strata   =         2                  Number of obs     =        3,658
Number of PSUs     =       100                  Population size   =  243,374,916
                                                Design df         =           98
                                                F(   4,     95)   =        20.24
                                                Prob > F          =       0.0000
```

seekcancerinfo	Coef.	Linearized Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	.3626786	.1345973	2.69	0.008	.0955746	.6297826
_Iedu_2	.8864881	.3021102	2.93	0.004	.2869604	1.486016
_Iedu_3	1.252377	.2894708	4.33	0.000	.6779312	1.826822
_Iedu_4	1.855047	.2904693	6.39	0.000	1.27862	2.431474
_cons	-1.646597	.3033351	-5.43	0.000	-2.248556	-1.044639

followed by all covariates. The **test** command tests the hypotheses about estimated parameters.

```
Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_2 = 0
 ( 3)  [seekcancerinfo]_Iedu_3 = 0
 ( 4)  [seekcancerinfo]_Iedu_4 = 0
 ( 5)  [seekcancerinfo]_cons = 0

       F(  5,    98) =   16.78
            Prob > F =    0.0000
```

```
Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_2 = 0
 ( 3)  [seekcancerinfo]_Iedu_3 = 0
 ( 4)  [seekcancerinfo]_Iedu_4 = 0

       F(  4,    98) =   20.88
            Prob > F =    0.0000

Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Igender_2 = 0

       F(  1,    98) =    7.26
            Prob > F =    0.0083

Unadjusted Wald test

 ( 1)  [seekcancerinfo]_Iedu_2 = 0
 ( 2)  [seekcancerinfo]_Iedu_3 = 0
 ( 3)  [seekcancerinfo]_Iedu_4 = 0

       F(  3,    98) =   25.33
            Prob > F =    0.0000
```

```
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running logit on estimation sample)

Survey: Logistic regression

Number of strata   =         2                  Number of obs      =        3,658
Number of PSUs     =       100                  Population size    =  243,374,916
                                                Design df          =           98
                                                F(   4,     95)    =        20.24
                                                Prob > F           =       0.0000
```

seekcancerinfo	Odds Ratio	Linearized Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	1.437174	.1934398	2.69	0.008	1.100291	1.877202
_Iedu_2	2.426593	.7330984	2.93	0.004	1.332371	4.419453
_Iedu_3	3.498648	1.012757	4.33	0.000	1.969798	6.214108
_Iedu_4	6.391998	1.856679	6.39	0.000	3.59168	11.37564
_cons	.1927045	.058454	-5.43	0.000	.1055515	.3518189

Note: **_cons** estimates baseline odds.

To identify levels/variables that display a significant difference in response, the rule of thumb is to examine odds ratios where the confidence interval does not contain 1 (by default, Stata will use alpha=.05 to determine statistical significance; the user can change this value by specifying a different threshold in the code). However, significance may also be garnered from the test of whether the associated beta parameter is equal to 0 (see first regression table above). According to this model, women appear to be 1.44 times as likely as men to have searched for cancer information.

Linear Regression

This example demonstrates a multivariable linear regression model using **svy: regress**; recall that the response should be a continuous variable. For the purposes of this example, we decided to use an outcome with five levels as a continuous variable (generalhealth). Note that higher values on generalhealth indicate poorer self-reported health status.

```
*   Multivariable linear regression of gender and education on generalhealth

xi: svy: regress generalhealth i.gender i.edu

test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4 _cons, nosvyadjust
test _Igender_2 _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
test _Igender_2, nosvyadjust
test _Iedu_2 _Iedu_3 _Iedu_4, nosvyadjust
```

```
i.gender          _Igender_1-2        (naturally coded; _Igender_1 omitted)
i.edu             _Iedu_1-4           (naturally coded; _Iedu_1 omitted)
(running regress on estimation sample)

Survey: Linear regression

Number of strata   =         2                  Number of obs      =        3,642
Number of PSUs     =       100                  Population size    =  242,654,074
                                                Design df          =           98
                                                F(   4,     95)    =        32.55
                                                Prob > F           =       0.0000
                                                R-squared          =       0.0741
```

generalhea~h	Coef.	Linearized Std. Err.	t	P>|t|	[95% Conf.	Interval]
_Igender_2	.0390436	.0453351	0.86	0.391	-.0509224	.1290096
_Iedu_2	.008683	.114848	0.08	0.940	-.2192291	.2365952
_Iedu_3	-.1848788	.1131293	-1.63	0.105	-.4093803	.0396227
_Iedu_4	-.6182037	.1012443	-6.11	0.000	-.8191198	-.4172877
_cons	2.779991	.1084629	25.63	0.000	2.56475	2.995232

```
Unadjusted Wald test

 ( 1)  _Igender_2 = 0
 ( 2)  _Iedu_2 = 0
 ( 3)  _Iedu_3 = 0
 ( 4)  _Iedu_4 = 0
 ( 5)  _cons = 0

       F(  5,    98) = 2824.23
            Prob > F =    0.0000

Unadjusted Wald test

 ( 1)  _Igender_2 = 0
 ( 2)  _Iedu_2 = 0
 ( 3)  _Iedu_3 = 0
 ( 4)  _Iedu_4 = 0

       F(  4,    98) =   33.58
            Prob > F =    0.0000
```

```
Unadjusted Wald test

 ( 1)  _Igender_2 = 0

       F(  1,    98) =    0.74
            Prob > F =    0.3912

Unadjusted Wald test

 ( 1)  _Iedu_2 = 0
 ( 2)  _Iedu_3 = 0
 ( 3)  _Iedu_4 = 0

       F(  3,    98) =   44.51
            Prob > F =    0.0000
```

From the above table, it can be seen that compared to those respondents with less than a high school education, those with a college degree or higher have a significantly negative linear association with the outcome (i.e., better reported health), controlling for all variables in the model. We don't interpret the gender variable because it is non-significant.

Merging HINTS Survey Iterations

This section provides SAS, SPSS, and Stata codes to combine HINTS 5, Cycle 4 and HINTS 5, Cycle 3 survey iterations. The provided code will generate one final sample weight for population point estimates and 100 replicate weights to compute standard errors.

Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using SAS

This section provides SAS (Version 9.3 and higher) code for merging the HINTS 5, Cycle 3 (2019) and HINTS 5, Cycle 4 (2020) iterations. HINTS 5, Cycle 3 employed a Web Pilot experimental design (more details can be found in the HINTS 5 Cycle 3 Methodology Report and HINTS 5 Cycle 3 Survey and Data Analysis Recommendations documents). It is suggested that analysts first assess for group differences in the HINTS 5, Cycle 3 data on the variables of interest (See "Assessing for Differences Across Groups" found in the Overview of the HINTS 5 Cycle 3 Survey and Data Analysis Recommendations document when you download the HINTS 5 Cycle 3 data for more information).

The code below is created assuming there are **no differences** between the different modalities in HINTS 5 Cycle 3 and uses the combined weights (TG_all weights). An analyst might find that there are group differences on the variables of interest in HINTS 5 Cycle 3. In this case, the analyst would need to create a new variable in both data files that would allow the analyst to differentiate between the 4 groups (HINTS 5 Cycle 4, HINTS 5 Cycle 3 [H5C3] Paper Only, H5C3 Web Option, and H5C3 Web Bonus groups). Next, the analyst will need to use the Rizzo, et al., (2008) method to create 200 replicate weights.

Alternatively, the analyst could use Taylor linearization to control for group differences by using the NWGT0 weight, VAR_STRATUM, and VAR_CLUSTER variables. In both the Rizzo method and the Taylor linearization method, analysts would need to control for group differences by using the created variable that allows the analyst to differentiate between the 4 groups.

An analyst may also choose to use only one group from Cycle 3 to combine with Cycle 4. One scenario is if an analyst decides they would like to keep modality consistent across both HINTS iterations, and thus it would be appropriate to use the "paper only" condition from Cycle 3 to combine with Cycle 4 data. This would be achieved by instead using the TG1 (instead of TG_all) variable weights from Cycle 3 in the below sample code.

One assumption when using the SAS code below is that the analyst has already formatted each file using the formats and format assignment files provided in the downloads.

In order to combine HINTS 5, Cycle 4 with HINTS 5, Cycle 3, the below sample code first creates a temporary format for a new "survey" variable that will distinguish between the two iterations. The code then creates two temporary data files and adds the new "survey" variable to each dataset. This survey variable can later be used to easily differentiate the cases that came from each HINTS iteration. Next, the two files are merged into one. It will match up variables that have the same name and format and create a merged data file (n = 9,303) that contains one final sample weight (for population point estimates; Merged_WGT0) and 100 replicate weights (Merged_WGT1 through Merged_WGT100; to compute standard errors).

```
/*FIRST CREATE THE FORMAT FOR THE SURVEY VARIABLE*/
proc format;
      value survey
      1="HINTS 5 CYCLE 3"
      2="HINTS 5 CYCLE 4"
      ;
run;
```

```
/*************************************************************************/

/*CREATE TWO SEPARATE TEMPORARY DATA FILES THAT CONTAIN THE NEW 'SURVEY'
VARIABLE.*/

/*PUT NAME OF LIBRARY WHERE HINTS 5 CYCLE 3 FORMATS ARE STORED*/;
options fmtsearch=(hints5c3);

data tempHINTS5CYCLE3;
      /*PUT NAME OF LIBRARY AND NAME OF EXISTING HINTS 5 CYCLE 3 DATA FILE*/
      set hints5c3.hints5_cycle3_public;
      survey=1;
      format survey survey.;
run;

/* PUT NAME OF LIBRARY WHERE HINTS 5 CYCLE 4 FORMATS ARE STORED*/
options fmtsearch=(hints5c4);

data tempHINTS5CYCLE4;
      /*PUT NAME OF LIBRARY AND NAME OF EXISTING HINTS 5 CYCLE 4 DATA FILE*/
      set hints5c4.hints5_cycle4_public;
      survey=2;
      format survey survey.;
run;

/*************************************************************************/
```

SAS Code to Set Up Final and Replicate Weights for the Replicate Variance Estimation Method

```
/*THIS CODE MERGES THE TWO TEMPORARY DATA SETS CREATED ABOVE. IT ALSO
CREATES ONE FINAL SAMPLE WEIGHT (Merged_NWGT0) AND 100 REPLICATE WEIGHTS
(Merged_NWGT1 THRU Merged_NWGT100)*/

options fmtsearch=(hints5c3 hints5c4);
data mergeHINTS5C3_HINTS5C4;
      set tempHINTS5CYCLE3 tempHINTS5CYCLE4;

      /*Create Replicate Weights for trend tests*/
      **Replicate Weights;
      array hints53wgts [50] TG_all_finwt1-TG_all_finwt50;
      array hints54wgts [50] person_finwt1-person_finwt50;
      array Merged_NWgt [100] Merged_NWGT1-Merged_NWGT100;

      **Adjust Final and Replicate Weights;
      if survey eq 1 then do i=1 to 50;   *HINTS 5 CYCLE 3;
            Merged_NWGT0=TG_all_finwt0;
            Merged_NWgt[i]=hints53wgts[i];
            Merged_NWgt[50+i]=TG_all_finwt0;
      end;

      else if survey eq 2 then do i=1 to 50; *HINTS 5 CYCLE 4;
            Merged_NWGT0=person_finwt0;
            Merged_NWgt[i]=person_finwt0;
            Merged_NWgt[50+i]=hints54wgts[i];
      end;
run;
```

```
/*********************************************************/
/*YOU CAN USE THE CODE BELOW TO RUN SIMPLE FREQUENCIES ON TWO COMMON
VARIABLES, 'SEEKCANCERINFO' AND 'CHANCEASKQUESTIONS'*/

/*SAS CODE*/
proc surveyfreq data = mergeHINTS5C3_HINTS5C4 varmethod = jackknife;
      weight Merged_NWGT0;
      repweights Merged_NWGT1-Merged_NWGT100 / df = 98 jkcoefs = 0.98;
      tables seekcancerinfo chanceaskquestions;
run;
```

SAS Code to Merge HINTS 5, Cycle 4 and HINTS 5, Cycle 3 for the Taylor Series Linearization Method

```
/*THIS CODE MERGES TWO TEMPORARY HINTS DATA SETS CREATED USING THE TAYLOR
SERIES LINEARZATION METHOD. PLEASE NOTE, THIS CODE IS BASED ON THE
ASSUMPTION THAT THE DATA SETS HAVE THE CORRECT VARIANCE CODES AND HHID
VARIABLES MATCH*/

/*FIRST CREATE THE FORMAT FOR THE SURVEY VARIABLE*/;
proc format;
      value survey
      1="HINTS 5 CYCLE 3"
      2="HINTS 5 CYCLE 4"
      ;
run;

/*********************************************************************/
/*CREATE TWO SEPARATE TEMPORARY DATA FILES THAT CONTAIN THE NEW
'SURVEY' VARIABLE AND BOTH CONTAIN THE SAME WEIGHT VARIABLES.*/
/* NOTE THAT IN THIS EXAMPLE WE USE THE TG_ALL_FINWT0 VARIABLE AS OUR
WEIGHTING VARIABLE FROM HINTS 5 CYCLE 3.
/*IF YOU WOULD LIKE TO USE A DIFFERENT WEIGHTING VARIABLE (E.G. YOU ARE
INTERESTED IN THE PAPER ONLY SAMPLE ONLY, TG1_FINWT0), YOU WILL NEED TO
UPDATE THE CODE IN THE HINTS5CYCLE3 RENAME LINE ACCORDINGLY*/

/*PUT NAME OF LIBRARY WHERE HINTS 5 CYCLE 3 FORMATS ARE STORED*/
options fmtsearch=(hints5c3);

data tempHINTS5CYCLE3;
      /*PUT NAME OF LIBRARY AND NAME OF EXISTING HINTS 5 CYCLE 3 DATA FILE*/
      set hints5c3.hints5_cycle3_public;
      RENAME TG_ALL_FINWT0=MERGED_FINWT0;

      survey=1;
      format survey survey.;
run;

/* PUT NAME OF LIBRARY WHERE HINTS 5 CYCLE 4 FORMATS ARE STORED*/
options fmtsearch=(hints5c4);
```

(code continued on the next page)

```
data tempHINTS5CYCLE4;
      /*PUT NAME OF LIBRARY AND NAME OF EXISTING HINTS 5 CYCLE 4 DATA FILE*/
      set hints5c4.hints5_cycle4_public;
      RENAME PERSON_FINWT0=MERGED_FINWT0;
      survey=2;
      format survey survey.;
run;

options fmtsearch=(hints5c3 hints5c4);

data mergeHINTS5C3_HINTS5C4;
      set tempHINTS5CYCLE3 tempHINTS5CYCLE4;
run;

/*********************************************************/
/*YOU CAN USE THE CODE BELOW TO RUN SIMPLE FREQUENCIES ON TWO
COMMON VARIABLES, 'SEEKCANCERINFO' AND 'CHANCEASKQUESTIONS'*/

/*SAS CODE*/
proc surveyfreq data = MergeHints5C3_Hints5c4 varmethod = TAYLOR;
      strata VAR_STRATUM;
      cluster VAR_CLUSTER;
      weight MERGED_FINWT0;
      tables seekcancerinfo chanceaskquestions / row col;
run;
```

Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using SPSS

This section provides SPSS (Version 25) syntax for merging the HINTS 5, Cycle 3 (2019) and HINTS 5, Cycle 4 (2020) iterations. Note that the below sample syntax is created with the assumption that there were no group differences found within HINTS 5, Cycle 3 and so we used the combined weights TG_ALL).

Within the below example SPSS syntax, a new "survey" variable is created in both datasets that will distinguish between the two iterations once the datasets are merged. Next, the two files are merged into one. It will match up variables that have the same name and format and create a merged data file (n = 9,303).

First, you will need to have **HINTS 5, Cycle 4** data open. The below syntax will first save a copy of HINTS 5, Cycle 4 and rename it as a new file called 'MERGED_H5C4_H5C3.sav. We highly suggest this step for several reasons, mainly being that when SPSS merges datasets the old file may be overwritten. By saving your original datafile, you can always have this available to refer to. Next, the syntax will rename the dataset to help with making sure the correct dataset is active and being edited in later syntax. Next, the below syntax copies Cycle 4's weighting variable PERSON_FINWT0 so that both cycles' weighting variable names match (MERGED_FINWT0). Finally, the syntax creates a new variable called 'Survey' and gives each participant in Cycle 4 a "2" so that analysts can easily identify cases from HINTS 5, Cycle 4.

```
SAVE OUTFILE='INSERT YOUR FILE PATH HERE\MERGED_H5C4_H5C3.sav'
 /COMPRESSED.
DATASET NAME MERGED_DATA.
```

(code continued on the next page)

```
DATASET ACTIVATE MERGED_DATA.
COMPUTE MERGED_FINWT0=PERSON_FINWT0.
COMPUTE Survey=2.
EXECUTE.
```

Next, we need to open our HINTS 5 CYCLE 3 data and rename our datafile, again to help with keeping files aligned for the merging process below. The following code will open your HINTS 5 Cycle 3 data and rename the dataset as H5C3. The syntax will then create the 'Survey' variable in the HINTS 5, Cycle 3 dataset and give each participant from Cycle 3 a value of "1". Again, this is so that once the datasets are merged, analysts can easily identify which cases were from the HINTS 5, Cycle 3 dataset. Finally, the syntax copies the weighting variable TG_ALL_FINWT0 and names it MERGED_FINWT0 so that the key weighting variable matches the key weighting variable from our HINTS 5 Cycle 4 dataset
Note, the analyst will need to insert the file path for where HINTS 5 Cycle 3 is saved.

```
**below, you should insert the filepath for your HINTS 5 Cycle 3 data**.
GET FILE='INSERT YOUR FILE PATH HERE\hints5_cycle3_public.sav'.
DATASET NAME H5C3 WINDOW=FRONT.
COMPUTE MERGED_FINWT0=TG_ALL_FINWT0.
COMPUTE Survey=1.
EXECUTE.
```

Next, a plan file is required to conduct analyses in SPSS. To create a plan file and subsequently conduct analyses, paste the following syntax in the SPSS Syntax Editor:

```
* Analysis Preparation Wizard.
*INSERT DATH OF PATH TO SAMPLE DESIGN FILE IN /PLAN FILE=.
CSPLAN ANALYSIS
 /PLAN FILE='INSERT YOUR FILE PATH HERE\MergePlan.csaplan'
 /PLANVARS ANALYSISWEIGHT=MERGED_FINWT0
 /SRSESTIMATOR TYPE=WOR
 /PRINT PLAN
 /DESIGN STRATA=VAR_STRATUM CLUSTER=VAR_CLUSTER
 /ESTIMATOR TYPE=WR.
```

Once you have your plan file, you can begin the merging process. You should, by this point, have two datasets open: "MERGED_H5C4_H5C3" (which currently contains only HINTS 5 Cycle 4's data) and "hints5_cycle3_public". Within your "MERGED_H5C4_H5C3" dataset you will navigate to the "Data" dropdown and select "Merge Files". You will be given the option to merge by cases or variables. Because we are merging two different cycles with mostly the same variables, we will want to select merge by "Add Cases". You will then select the hints5_cycle3_public dataset that is open from the window that pops up and click continue. Ensure that the variables you need in the new merged dataset you are creating are in the "Variables in New Active Dataset" box. Once you have verified all your desired variables are in that box, click "OK".

```
DATASET ACTIVATE MERGED_DATA.
ADD FILES /FILE=*
  /FILE='H5C3'.
EXECUTE.
```

(code continued on the next page)

```
/*****************************************************/

**YOU CAN USE THE CODE BELOW TO RUN SIMPLE FREQUENCIES ON TWO COMMON
VARIABLES, 'seekcancerinfo' AND 'chanceaskquestions'*/

*INSERT PATH OF TO ANALYSIS PLAN UNDER /PLAN FILE.
CSTABULATE
/PLAN FILE='INSERT YOUR FILE PATH HERE\MergePlan.csaplan'
/TABLES VARIABLES=seekcancerinfo chanceaskquestions
/CELLS POPSIZE TABLEPCT
/STATISTICS SE COUNT
/MISSING SCOPE=TABLE CLASSMISSING=EXCLUDE.
```

Merging HINTS 5, Cycle 4 and HINTS 5, Cycle 3 using Stata

This section provides Stata (Version 10.0 and higher) code for merging the HINTS 5, Cycle 3 (2019) and HINTS 5, Cycle 4 (2020) iterations. HINTS 5, Cycle 3 employed a Web Pilot experimental design (more details can be found in the HINTS 5 Cycle 3 Methodology Report and HINTS 5 Cycle 3 Survey and Data Analysis Recommendations documents). It is suggested that analysts first assess for group differences in the HINTS 5, Cycle 3 data on the variables of interest (See "Recommendations for Statistical Analyses Using HINTS 5 Cycle 3: Assessing for Differences Across Groups" for more information).

The code below is created assuming there are **no differences** between the different modalities in HINTS 5 Cycle 3 and uses the total weights (TG_all weights). An analyst might find that there are group differences on the variables of interest in HINTS 5 Cycle 3. In this case, the analyst would need to create a new variable in both data files that would allow the analyst to differentiate between the 4 groups (HINTS 5 Cycle 4, HINTS 5 Cycle 3 [H5C3] Paper Only, H5C3 Web Option, and H5C3 Web Bonus groups). Next, the analyst will need to use the Rizzo, et al., (2008) method to create 200 replicate weights.

Alternatively, the analyst could use Taylor linearization to control for group differences by using the NWGT0 weight, VAR_STRATUM, and VAR_CLUSTER variables. In both the Rizzo method and the Taylor linearization method, analysts would need to control for group differences by using the created variable that allows the analyst to differentiate between the 4 groups.

An analyst may also choose to use only one group from Cycle 3 to combine with Cycle 4. One scenario is if an analyst decides they would like to keep modality consistent across both HINTS iterations, and thus it would be appropriate to only use the "paper only" condition from Cycle 3 to combine with Cycle 4 data. This would be achieved by instead using the TG1 (instead of TG_all) variable weights from Cycle 3 in the below sample code.

Stata Code to Set Up Final and Replicate Weights for the Replicate Variance Estimation Method

In order to combine HINTS 5, Cycle 3 with HINTS 5, Cycle 4, the below sample code creates two temporary data files and generates the appropriate final sample weight (for population point estimates; Merged_NWGT0) and 100 replicate weights (Merged_NWGT1 through Merged_NWGT100; to compute standard errors) on each, using the Rizzo, et al., (2008) method. Next, the two files are merged into one and the new "survey" variable is generated to distinguish between the two iterations. This survey variable can later be used to easily differentiate the cases that came from each HINTS iteration. During the merge, Stata will match up variables that have the same name and format, creating a final merged data file (n = 9,303).

```
*Put path and name to your HINTS 5 Cycle 3 data
use "INSERT YOUR PATH HERE\hints5_cycle3_public.dta", clear

*Create final and replicate weights (merged_nwt*) for multi-cycle datasets
gen merged_nwgt0=tg_all_finwt0
forvalues n1=1/50 {
local x1=`n1'+50
gen merged_nwgt`n1'=tg_all_finwt`n1'
gen merged_nwgt`x1'=tg_all_finwt0
}
save h5c3.dta, replace

*Put path and name to your HINTS 5 Cycle 4 data
use "INSERT YOUR PATH HERE\hints5_cycle4_public.dta", clear
*Create final and replicate weights (merged_nwt*) for multi-cycle datasets
gen merged_nwgt0=person_finwt0
forvalues n2=1/50 {
local x2=`n2'+50
gen merged_nwgt`n2'=person_finwt0
gen merged_nwgt`x2'=person_finwt`n2'
}
save h5c4.dta, replace

set trace off

*Combine the 2 cycles of data & generate survey variable flagging HINTS
iteration
use h5c3.dta, clear
append using h5c4.dta, generate(survey)
label define survey 0 "HINTS 5 CYCLE 3" 1 "HINTS 5 CYCLE 4"
label values survey survey
save combined.dta, replace

* Use the code below to run simple one-way frequencies for 2 common variables
** First, declare survey design
svyset [pw=merged_nwgt0], jkrw(merged_nwgt1-merged_nwgt100, multiplier(0.98))
vce(jack) dof(98) mse

svy: tabulate seekcancerinfo, obs percent se
svy: tabulate chanceaskquestions, obs percent se
```

Stata Code to Merge HINTS 5, Cycle 4 and HINTS 5, Cycle 3 for the Taylor Series Linearization Method

In order to combine HINTS 5, Cycle 3 with HINTS 5, Cycle 4, the below sample code creates two temporary data files and generates the appropriate final sample weight (for population point estimates; Merged_NWGT0) on each. No transformations are needed to the VAR_CLUSTER and VAR_STRATUM variables to support computation of standard errors. Next, the two files are merged into one and the new "survey" variable is generated to distinguish between the two iterations. This survey variable can later be used to easily differentiate the cases that came from each HINTS iteration. During the merge, Stata will match up variables that have the same name and format, creating a final merged data file (n = 9,303).

```
*Put path and name to your HINTS 5 Cycle 3 data
use "INSERT YOUR PATH HERE\hints5_cycle3_public.dta", clear
```

(code continued on the next page)

```
*Create final weight (merged_nwt0) for multi-cycle datasets
gen merged_nwgt0=tg_all_finwt0
save h5c3.dta, replace

*Put path and name to your HINTS 5 Cycle 4 data
use " INSERT YOUR PATH HERE\hints5_cycle4_public.dta", clear
*Create final weight (merged_nwt0) for multi-cycle datasets
gen merged_nwgt0=person_finwt0
save h5c4.dta, replace

*Combine the 2 cycles of data & generate survey variable flagging HINTS
iteration
use h5c3.dta, clear
append using h5c4.dta, generate(survey)
label define survey 0 "HINTS 5 CYCLE 3" 1 "HINTS 5 CYCLE 4"
label values survey survey
save combined.dta, replace

* Use the code below to run simple one-way frequencies for 2 common variables
** First, declare survey design
svyset var_cluster [pw=merged_nwgt0], strata(var_stratum)
svy: tabulate seekcancerinfo, obs percent se
svy: tabulate chanceaskquestions, obs percent se
```

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